CS-011 Inventory, Storage & Transport

Doc ID	CS-011
Title	Inventory, Storage & Transport
Revision	B
Status	Draft, pending Lead signature (storage-map locations to field-verify)
Effective date	2026-07-28
Supersedes	The stale Master Tracker Inventory sheet as a live record

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted with Claude, SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial outline
B	2026-07-28	Claude	Added a reference to the app's Stock tab and noted its location field is freeform today, not a controlled vocabulary, so it doesn't yet enforce the storage map. Language pass.

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1. Purpose

Our SN5 inventory sheet was written at season start and then never touched. MEKP sat flagged "RE-ORDER" for months with no owner and no trigger, and when the October Easy Composites order got stuck in customs we had no buffer and lost weeks of layups (PP-02). Storage was purely reactive: parts got evicted from the etch locker, molds blocked everyone's boxes at RFS (PP-10). This doc sets minimum stocks, gives them an owner, and maps where things live.

2. Scope

Consumables, resins/chemicals, cloth, tooling board, molds, and finished parts: where they live, when they get reordered, how they move.

3. References

Ref	Document	Where
R1	PP-02, PP-10 RCAs	FEB-COMP-PP-001
R2	CS-012 (ordering process + lead times)	CS-012
R3	CS-001 (labels carry home locations)	CS-001
R4	SDS chemical storage sections	`04 Datasheets/`
R5	The app's Stock tab (live tooling-board + offcut inventory)	`03 App/app/README.md`

4. Definitions

Min-stock (the level that triggers a reorder — sized so the order lands before stock-out at the supplier's lead time), stock walk, home location.

5. Equipment & Materials — layup-blocker min-stock list (starting values; tune with usage data)

Sizing rule (§4): min stock = (burn rate × supplier lead time) + one job's worth. Worked examples for the three items that actually stop a layup — the rest of the table starts as one-job buffers, tuned with WO usage data:

- **IN2 + AT30:** SN5 logged ~1.95 kg mixed per large infusion; at the 2-infusions/week peak that is ~4 kg/week. Easy Composites lead 6 weeks (CS-012) → running dry mid-season costs ~6 weeks of infusions. One unopened kit (~4 kg) 1 peak week: it is the *alarm*, not the buffer — when the walk finds only the unopened kit, the order is already late, so trigger at **opened kit + 1 unopened**, and place the big seasonal orders per CS-012 §7.4 so the walk never trips this.
- **XCR:** ~1 kit seals 2–3 mold sections (tune from WO records); molds queue 2/week in peak → 1 kit 1 week. Same logic: **1 unopened kit minimum**, ordered with every EC basket.
- **Tacky tape:** every bagged job uses 1–2 rolls of perimeter; 2 jobs/week → ~3 rolls/week; domestic lead ~1 week → **6 rolls** 2 weeks of peak burn.

Item	Min stock	Why it blocks
IN2 resin + AT30	1 unopened kit (+ the opened one)	Every infusion; 6-week EC lead
West 105 + 206	½ tank + 1 can	Every wet layup
XCR	1 kit	Every mold
VB160 bag film	1 roll	Everything bagged
Peel ply / flow mesh	1 roll each	Everything bagged
Tacky tape	6 rolls	Everything bagged; team burns it fast
Nitrile gloves	2 boxes/size	Everything
Mixing cups/sticks	50/50	Everything
Respirator cartridges (A + P100)	2 sets	Safety-blocking
195 twill cloth	(sponsor) 10 yd	Most stacks

6. Safety

Chemical storage per each SDS (R4): resins/hardeners in the chemical boxes/barrel, **never in food fridges** (2026-02-17 incident); flammables (Airtac, acetone, clear coat) in the flammables cabinet; segregate hardeners from resins on different shelves.

7. Procedure (outline)

1. **Monthly stock walk (15 min, named owner — an IA/first-year is ideal):** check §5 list against reality; anything at/below min goes to #purchasing the same day per CS-012.
2. **Order timing:** Easy Composites 6 weeks before need (UK + customs, PP-02); one big fall order + one spring top-up beats four rushed baskets.
3. **Storage map (fill in real locations on first revision):** molds → RFS storage container (never in front of the knaack/screw boxes); finished parts → Jacobs basement shelves (never the etch locker — 2026-04-20 agreement); cloth → dry sealed bin; chemicals → per §6; consumables → General Box at RFS. Board and offcuts also get logged in the app's Stock tab (R5), which is where real inventory now lives day to day. As of 2026-07-28 its location field is freeform text (placeholder: "e.g. rack A, top shelf"), not a controlled list, so it can't yet enforce the storage map above; type the real location string consistently until it can.
4. **Transport:** molds over ~1 m^2 = two people + a vehicle that actually fits them (budget shows a U-Haul was needed in SN5 — plan it, don't discover it); parts travel padded and labeled.
5. **Handoff:** at season end, the stock walk owner + lead produce a one-page "what we actually have" list for the incoming lead (the 6/15 handoff gap).

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Stock walk done	Monthly during build season	Owner posts result in #composites	Slack + walk sheet
No layup blocked by stock-out	0 per season	WO delay notes	WO timeline
Molds in home locations	100% at walk	Walk	Walk sheet + WO mold.location

9. Records

Walk sheets (photo → Drive), WO mold.location fields, reorder requests in #purchasing.